Professional Certificate of Competency in Electrical Power System Fundamentals for Non-Electrical Engineers

Webinar 2

Module 2: Electrical Generation, Transmission and Distribution

Module 2 Topics

- Various forms of energy and energy conversion
- Conventional power sources: coal fired, hydro and nuclear power plants
- Renewable power sources: solar, wind, biomass
- Transmission and distribution of electricity

What is Energy?

Energy is the capacity to do work.

- **Energy = Power x Time** (Power is energy consumed per unit of time)
  - Energy can be used to move/lift objects (Mechanical energy)
  - Energy can raise the temperature of a substance (Heat energy)
  - Energy can produce light (Ex: LED, Fluorescent lighting)
  - Energy can cause chemical reactions (Ex: Electrolysis)
  - Energy is broadly divided into two main categories

Energy Forms and Conversions

Energy:

- **Potential energy :**
  - Stored type of energy
    - Gravitational
    - Mechanical Energy
    - Chemical Energy
    - Nuclear Energy
- **Kinetic energy:**
  - Form of movement
    - Motion
    - Thermal Energy
    - Radiant Energy
    - Sound
    - Electricity

Electric Energy

- Probably the *most flexible* energy form.
- Can be generated from a variety of other energy sources: thermal, wind, solar etc.
- Can be converted back to other energy forms: heat, light, motion, etc.
- Has relatively low losses in long distance transmission, distribution and utilisation.
- Modern society is highly dependent on it.

Energy Sources

Primary sources:

- Renewable sources: An energy source that are replenished naturally in a short period of time. Examples: wind energy, solar energy, water energy, biomass, geothermal energy.
- Non-renewable sources: An energy source that we are using up and cannot get recreated in a short period of time. Examples: nuclear energy, natural gas, coal, oil.

Secondary sources:

- Examples are electricity and hydrogen of which Electricity can be generated in many ways.

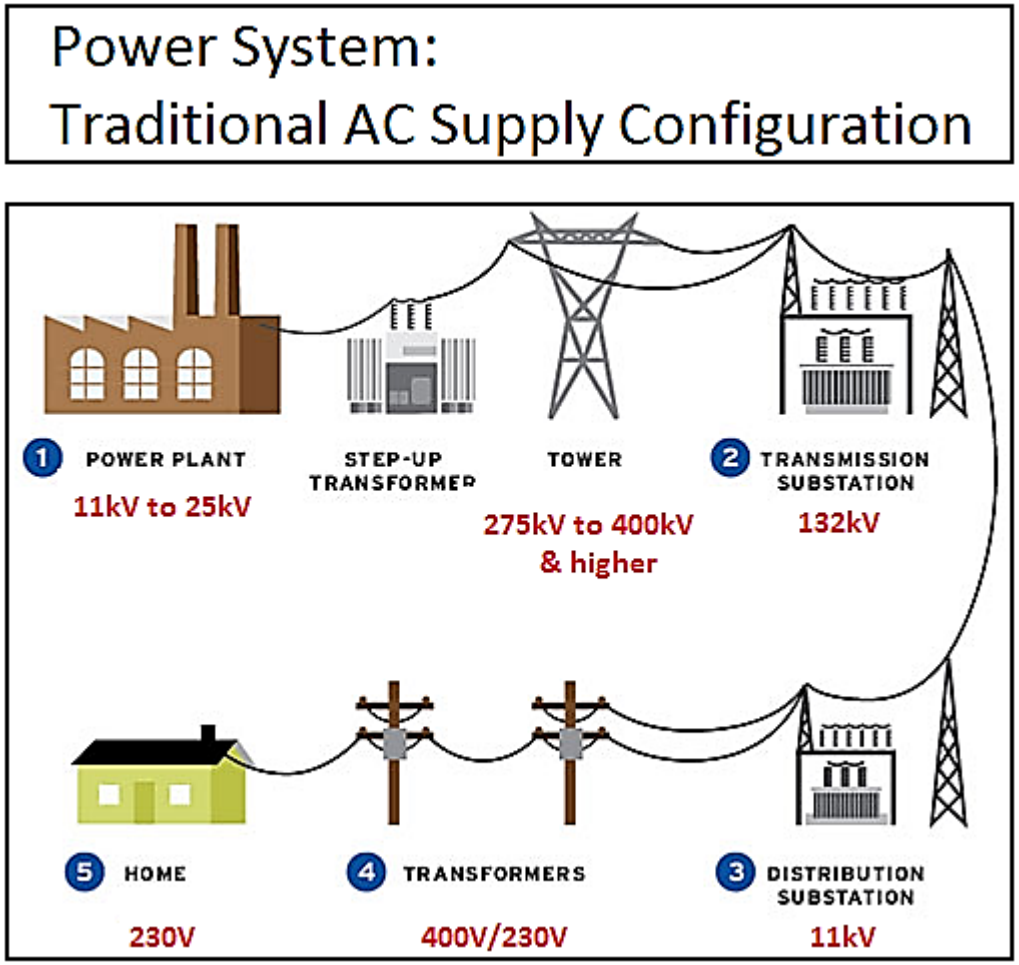
Types of Generators

- Hydro
- Pump storage
- Wind
- Solar
- Geo-thermal
- Tidal wave
- Coal fired
- Nuclear
- Gas – aircraft derivative
- Micro-turbine
- Combined cycle gas
- Diesel

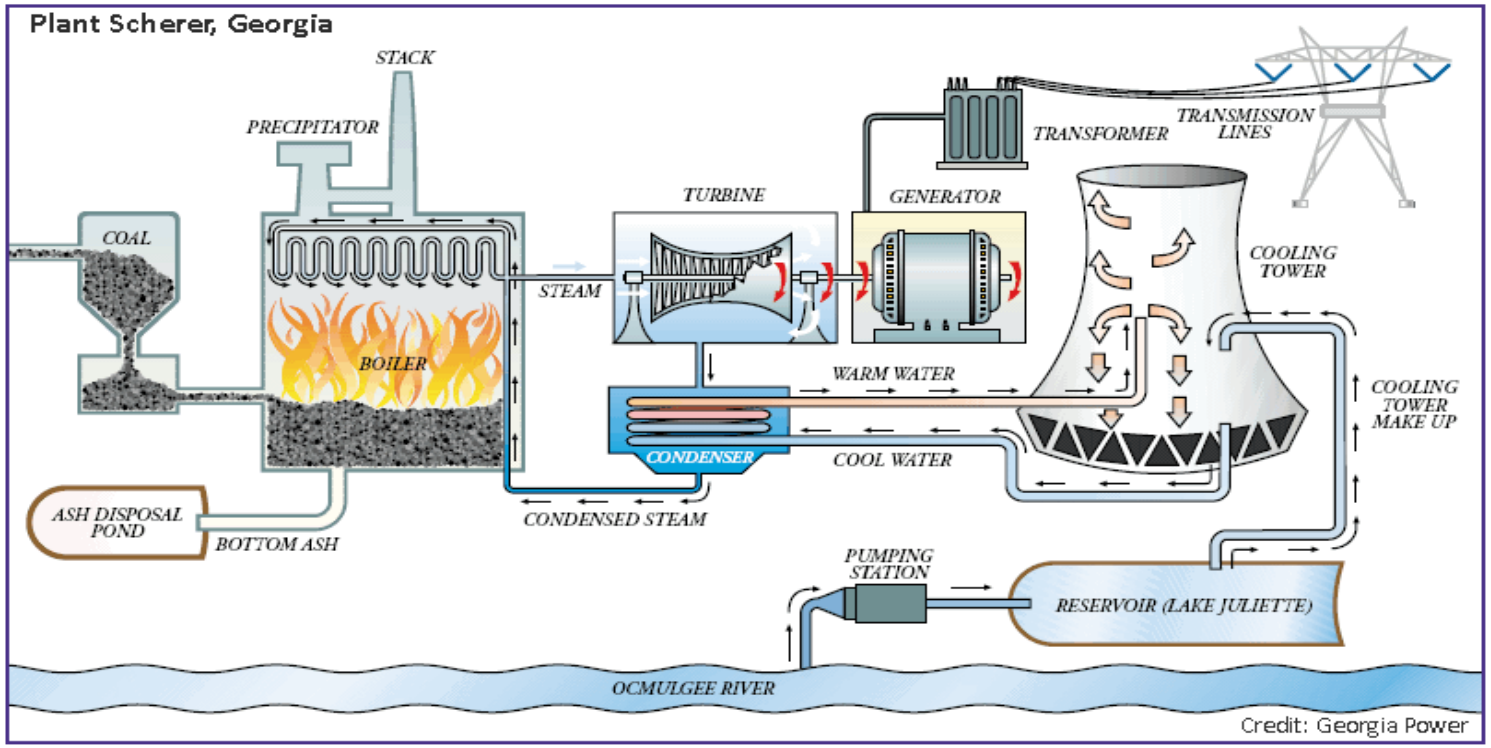
Basis of Voltage Selection

- **System segment:**
  - Generation: MV
  - Transmission: HV and Extra High Voltage (230kV+)
  - Sub-transmission: HV/MV
  - Distribution: MV/LV
  - Utilization: MV (Industries)/LV (Domestic, Small units)
- The sizes of equipment adopted differ in the above systems based on the magnitudes of electrical parameters like operating voltage, power to be handled, etc., calling additional space and protection functions as the voltage levels and power go up.

Traditional Power Supply Configuration



Coal-fired Power Plant

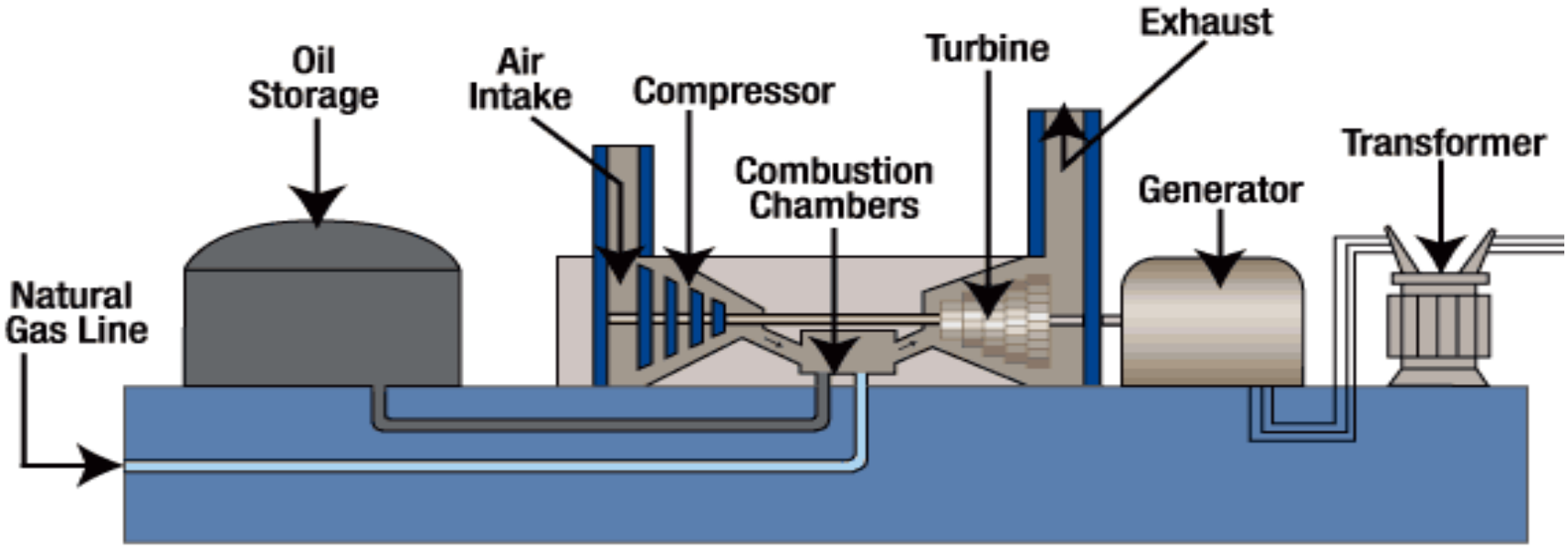


Burn Fuel → heat water to make steam → steam turns turbines → turbines turn generators → electrical power sent around country

Plant location is determined by closeness to source of fuel and water source

https://www.usgs.gov/special-topic/water-science-school/science

Gas-fired Power Plant



Burn Fuel → hot gases expand through turbine blades → turbines turn generators → electrical power sent around country

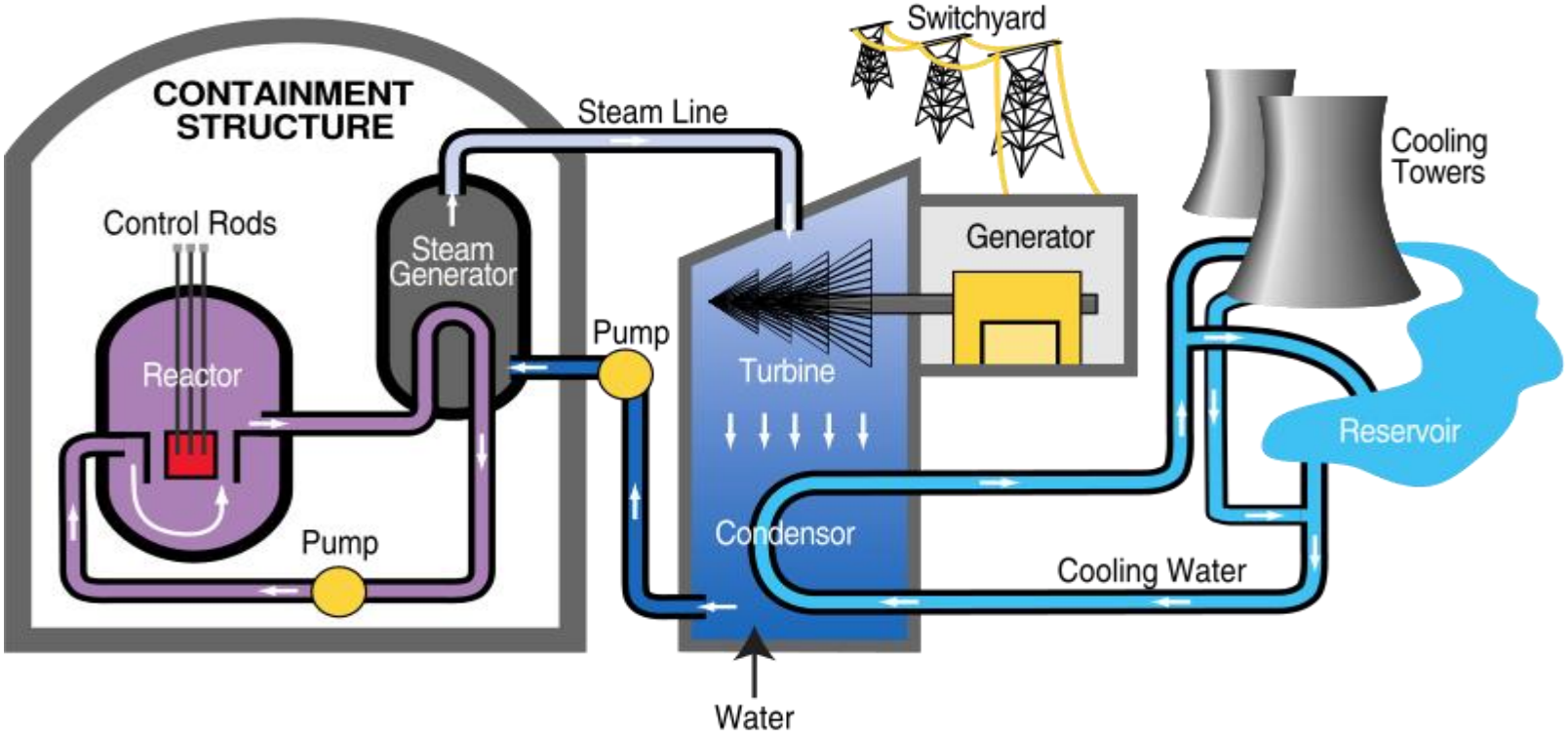
https://www.tva.com/energy/our-power-system/natural-gas/how-a-combustion-turbine-plant-works

Combined Cycle with Cogeneration

This principle is adopted in power plants and industries which employ gas turbines.

- Exhaust gases from the main turbine carry a lot of heat
- This heat can also be used to produce steam
- The steam so produced runs a steam turbine to produce electricity
- Called *combined cycle generation*
- Efficiencies of up to 60%

Nuclear Power Plant



Nuclear fission makes heat → heat water to make steam → steam turns turbines → turbines turn generators → electrical power sent around country

https://en.wikipedia.org/wiki/File:PWR_nuclear_power_plant_diagram.svg

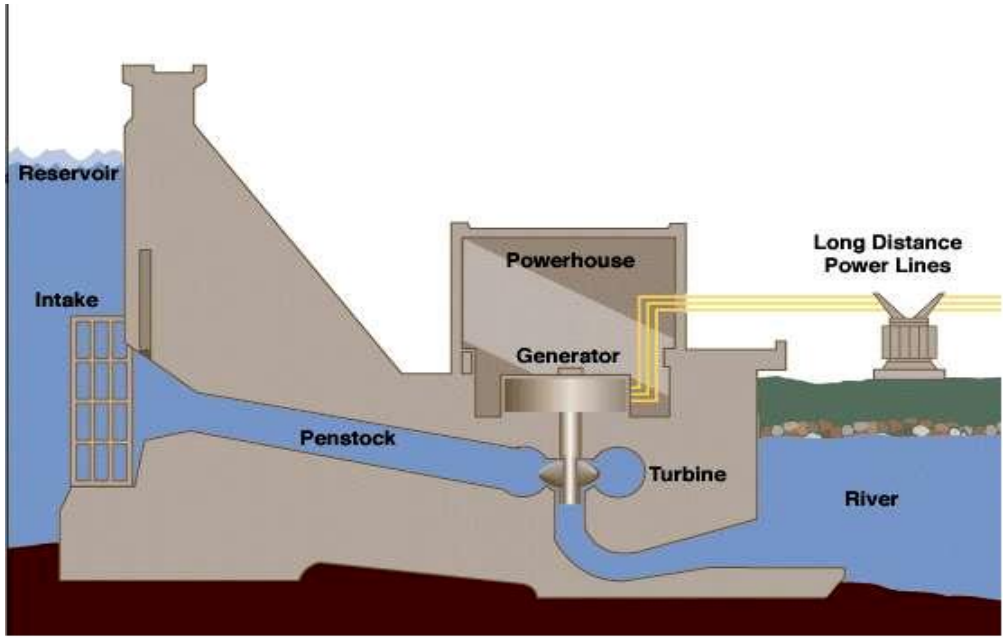
Renewable Energy

Renewable energy sources are alternative sources to generate electricity derived from natural resources that get replenished naturally.

- Hydroelectric power
- Solar energy
- Geothermal energy
- Biomass energy
- Ocean energy
- Wind energy

Hydro-electric Power Plant

- In hydro-electric power plants, gravity causes the water to flow and the kinetic energy contained in falling water generates electricity.
- Water returns back to reservoir and ensures continuous power production.



https://www.tva.com/energy/our-power-system/hydroelectric/how-hydroelectric-power-works

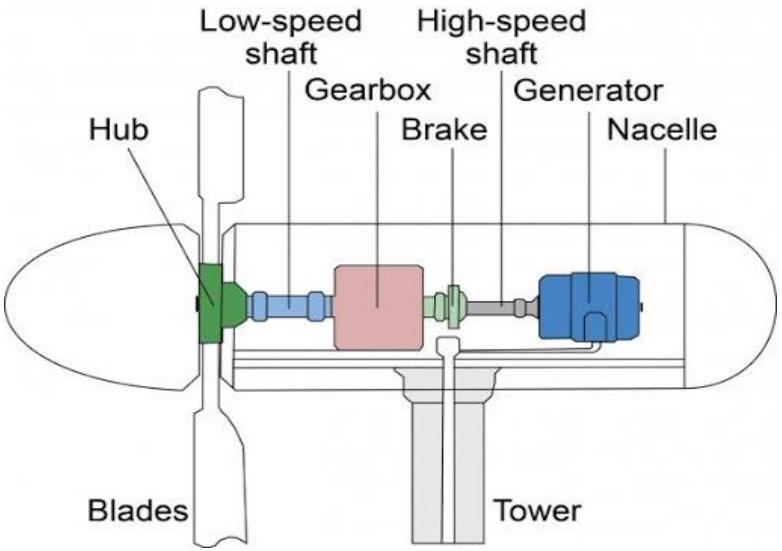
Solar Energy

- Sunlight provides energy which can be used to provide heat and generate electricity using photovoltaic effect.
- There are **two kinds** of *solar panels:*

1. Solar thermal panels, which heat water (without conversion to electricity).
2. Solar PV panels, which change light into electricity (dc) for power distribution.

Wind Energy

- When the wind blows its kinetic energy turns the turbine blades.
- Turbine converts wind energy into mechanical energy.
- Generator converts the rotational mechanical energy into electrical energy (seasonal).



http://css.umich.edu/factsheets/wind-energy-factsheet

Biomass

Biomass is any plant, animal or organic waste material which are found to contain high power gases like methane, etc.

- This can used as a fuel to provide thermal energy.
- The thermal output can be used directly for heating applications or to make steam for generating electricity.

Location of Power Plants

- Need to be located close to an energy resource
- Need to be close to power infrastructure to feed power to the grid

- These can sometimes be in conflict and economics will determine optimal location

Other major issues that need to be considered include:

- Safety (especially nuclear)
- Waste generation
- Availability of other resources (skills, water etc.)
- Environmental impact
- Sustainability

Generation and Transmission Voltages

- Electricity is generated at a voltage between **11 kV and 26 kV**.
- Depending on the distances involved it is then transformed up to a suitably higher value.
- Transmission takes place at very high voltages to **minimize losses** (132 kV or more**), reduce voltage drop and conductor size- all good advantages**
- The transmission lines are interconnected into the electric grid.
- The grid feeds electricity to substations where the voltage can be transformed to lower values.
- Power transmission is between power station and substations at different locations.

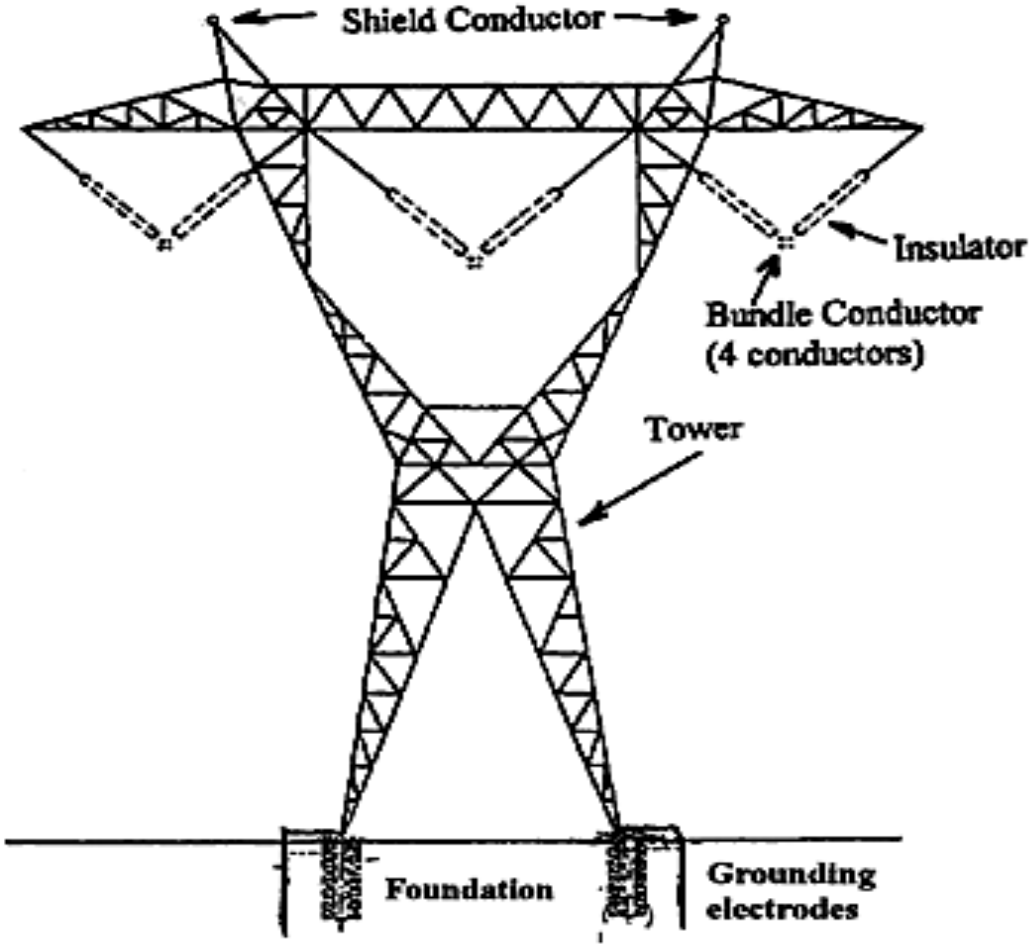
Electric Power Transmission

- Transmission is carried out by bare overhead conductors strung between transmission line towers to support the overhead lines with insulators.
- Transmission can also take place using underground power cables in urban situations.

Transmission structures include:

- Steel towers or poles
- Concrete poles
- Wooden poles (older sub-transmission structures)

Transmission Tower



Cable Distribution

- Used where exposed line poses hazards due to proximity
- Utilities use cables in a central business district distribution
- Industries invariably use cables for internal distribution
- Cables available for a wide range of voltages
- Installation and troubleshooting involve greater effort, time and cost
- Less frequent failures and lower maintenance

HVDC

- Voltage in ac can be increased or decreased with a transformer while dc needs power electronics components (inverter and rectifier) to transform voltages from one value to another.
- AC power transmission over long lengths is affected by inductance (L) and capacitance (C) parameters. However, dc power distribution is affected only by resistance but not by L or C since frequency = 0.
- HVDC enables bulk power transmission over very long distances, with higher efficiency and lower electrical losses per 1,000 km due to the absence of L and C.
- In a long ac cable transmission, the reactive power flow due to the large cable capacitance will limit the maximum possible transmission distance.
- With HVDC there is no such limitation; this is why, for very long cable links, HVDC is a viable technical transmission alternative.

Sub-transmission Systems

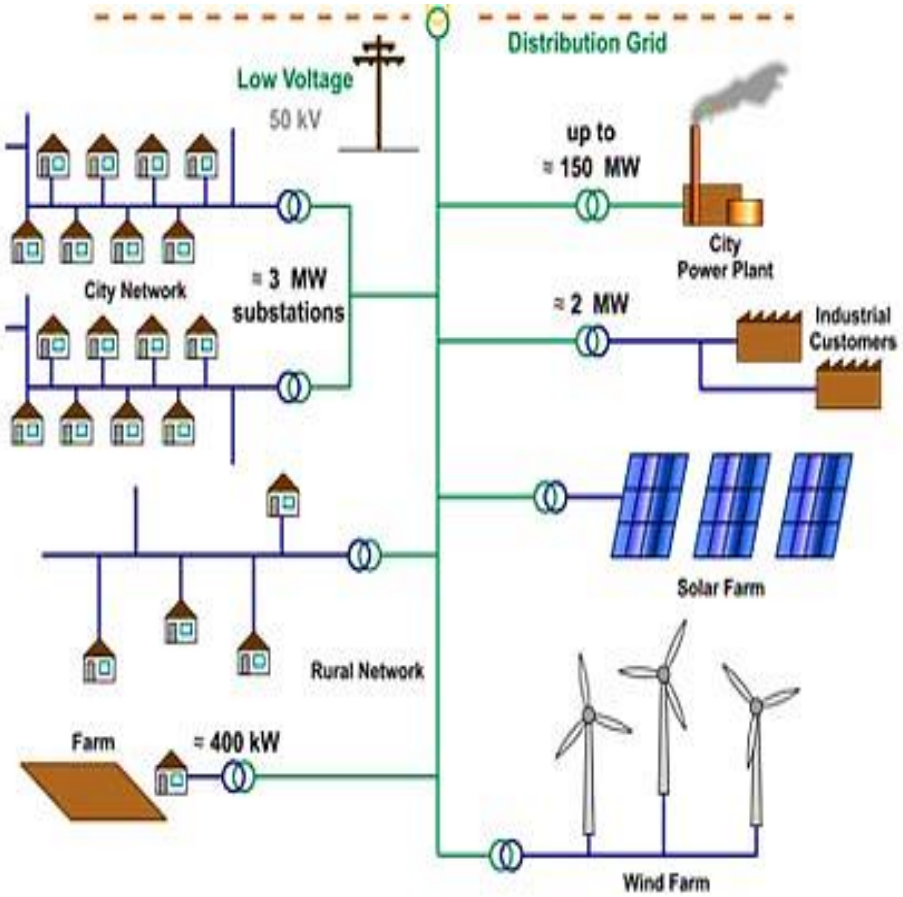
- Sub-transmission systems are connected to the bulk supply and supply distribution substations.
- A sub-transmission systems can supply several distribution substations.
- Common sub-transmission voltages range from 33 kV to 132 kV.

Electrical Distribution Network

- Customer equipment cannot utilise the high transmission or sub-transmission voltages so it has to be reduced and distributed.
- Large bulk consumers such as steel mills, petrochemical plants and mines will connect to the grid a high voltage and then transform and distribute around the plant at medium voltage (e.g. 33 kV).
- Smaller consumers and commercial establishments connect to a distribution network which is a medium (e.g. 11 kV) or low voltage (400 V three phase or 230 V single phase) depending on their power loads.
- Pole mount and pad mount arrangements for distribution transformers used to supply residential customers.

Distributed Generation

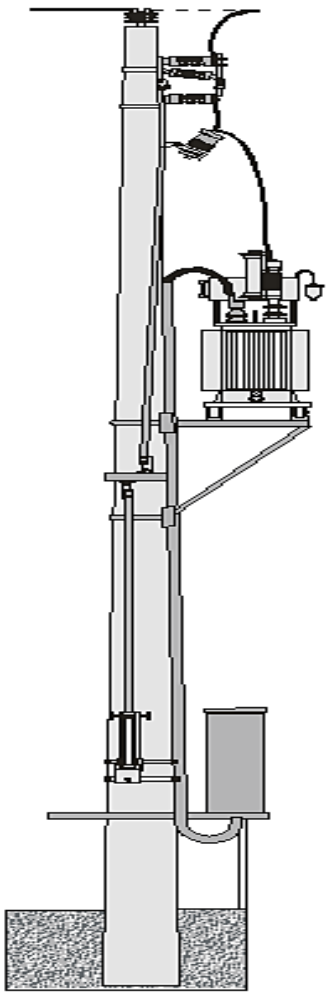
- Increasing use of renewable energy is changing grids from centralised to decentralised.
- Renewables provide local power generation which can be fed in at the distribution level close to the consumer.
- This often eliminates the need for expensive transmission infrastructure.



https://conduitnw.org

Pole-mounted Transformers

- Generally of smaller capacity single-phase and three-phase transformers.
- Primary protected with drop out fuses and surge arrestors.
- Economical and widely used for rural/remote power distribution.



Padmounted Transformers

- More popular in urban situations.

Overhead Distribution Lines

End of Webinar 2

Any questions or comments?